

07

Scientists investigated biodiversity in prokaryote communities found in soil.

The scientists:

- took soil samples from fields that had been managed for 20 years with two different farming methods
- sequenced all the DNA that coded for prokaryotic ribosomal RNA in the soil samples
- compared these base sequences to give a measure of species richness and an index of diversity for the prokaryote community
- recorded the total prokaryotic biomass and the mass of stored carbon for each soil sample
- obtained the mean wheat yield from the fields.

Table 4 shows the scientists' results.

Table 4

Data collected	Farming method 1	Farming method 2
Mean species richness ($\pm 2 \times$ standard deviation)	517 (± 17)	560 (± 24)
Mean index of diversity ($\pm 2 \times$ standard deviation)	0.251 (± 0.011)	0.230 (± 0.014)
Mean total prokaryotic biomass / kg m^{-3}	0.24	0.40
Mean carbon stored in soil organisms / $\mu\text{g g}^{-1}$	203	342
Mean wheat yield / g m^{-2}	451	377

The mean $\pm 2 \times$ standard deviation includes 95% of the data.

Do not write
outside the
box



07.2

Genetic diversity in soil species was traditionally inferred by making observations after growing prokaryotes on agar plates.

However, it is estimated that less than 10% of prokaryotes found in soil will grow if spread on an agar plate in a laboratory.

In recent years, our knowledge of prokaryotic biodiversity in the soil has increased.

Suggest why.

[2 marks]

07.3

Evaluate the balance between conservation and farming for these two farming methods.

Use the information provided in **Table 4** (on page 16).

[2 marks]

